

Take $d\vec{s}$ as given.

Shortest is $L = \int_{s_1}^{s_2} d\vec{s}$

R is constant so $dR = 0$.

2.

$d\vec{s} = |ds| d\hat{s}$

So, $|ds| = \sqrt{(Rd\theta)^2 + (Rd\phi \sin\theta)^2} = \sqrt{R^2(d\theta^2 + d\phi^2 \sin^2\theta)} = R d\theta \sqrt{1 + (\sin^2\theta) \left(\frac{d\phi}{d\theta}\right)^2}$

Thus,

$L = \int_{s_1}^{s_2} d\vec{s} = R \int_{\theta_1}^{\theta_2} d\theta \sqrt{1 + \sin^2\theta \left(\frac{d\phi}{d\theta}\right)^2}$

by property of Euler-Lagrange eq.

b.

Take $d\vec{s}$ as given, same thing, $R = \rho$ is constant

$L = \int_{s_1}^{s_2} d\vec{s} \quad d\rho = 0$

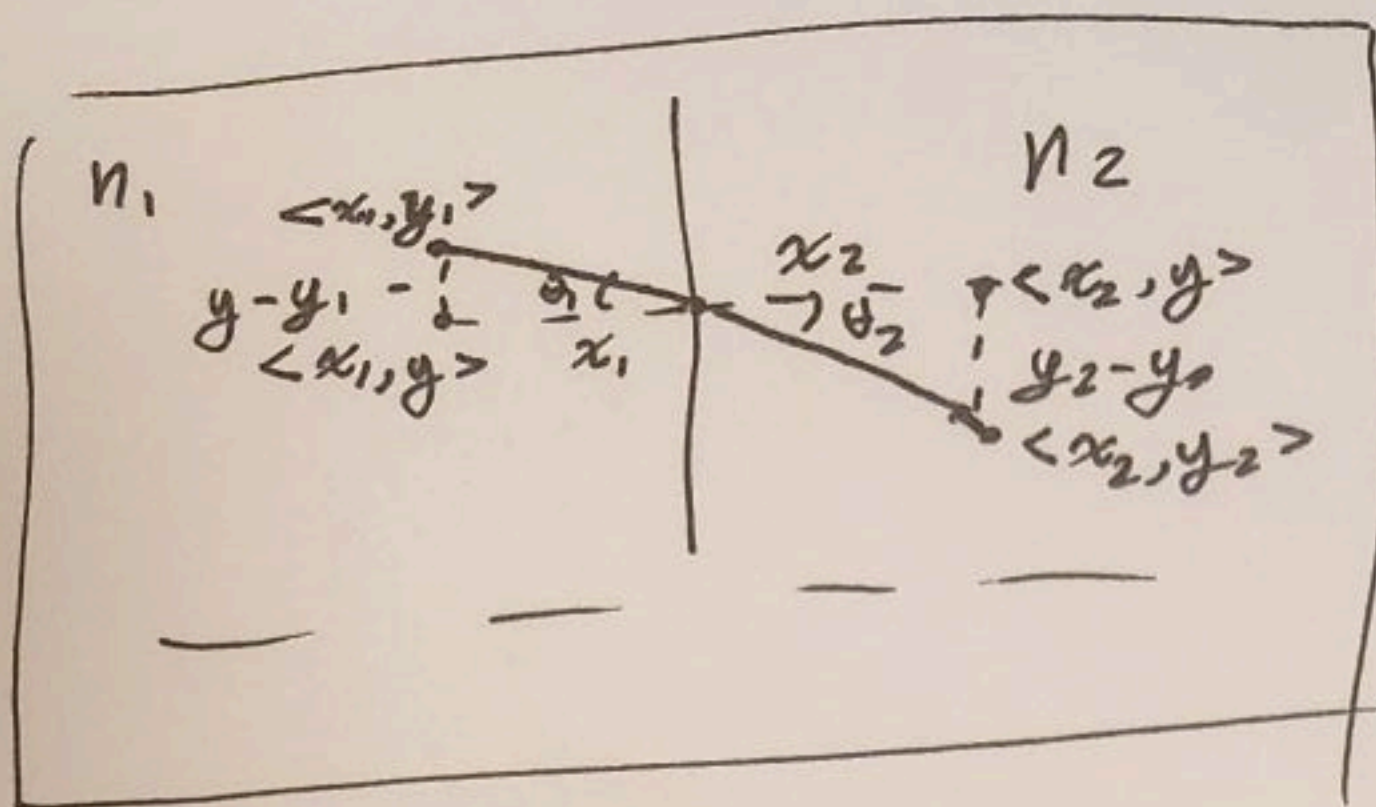
$d\vec{s} = |ds| d\hat{s} = \sqrt{(\rho d\phi)^2 + (dz)^2} = dz \sqrt{\rho^2 \left(\frac{d\phi}{dz}\right)^2 + 1}$

$\therefore L = \int_{z_1}^{z_2} dz \sqrt{\rho^2 \left(\frac{d\phi}{dz}\right)^2 + 1}$



#2.

2.



$$\cot \theta_1 = \frac{x_1}{(y-y_1)}$$

$$\cot \theta_2 = \frac{x_2}{(y-y_2)}$$

$$(b/c) \quad T = t_1 + t_2 = \frac{L_1}{v_1} + \frac{L_2}{v_2} = \frac{n_1 L_1}{c} + \frac{n_2 L_2}{c}$$

$$= \frac{n_1}{c} \cdot \sqrt{x_1^2 + (y-y_1)^2} + \frac{n_2}{c} \sqrt{x_2^2 + (y-y_2)^2}$$

~~$$= \frac{n_1}{c} \sqrt{(y-y_1)^2 \left(\frac{x_1^2}{(y-y_1)^2} + 1 \right)} + \frac{n_2}{c} \sqrt{(y-y_2)^2 \left(\frac{x_2^2}{(y-y_2)^2} + 1 \right)}$$~~

we pull out a negative one from $(y_2 - y)$!

$\frac{\partial T}{\partial y} \stackrel{!}{=} 0$, we evaluate.

$$0 = \frac{\partial T}{\partial y} = \frac{n_1}{c} \left(\frac{y-y_1}{\sqrt{x_1^2 + (y-y_1)^2}} \cdot \frac{1}{\sqrt{\frac{x_1^2}{(y-y_1)^2} + 1}} \right) - \frac{n_2}{c} \left(\frac{y-y_2}{\sqrt{x_2^2 + (y-y_2)^2}} \cdot \frac{1}{\sqrt{\frac{x_2^2}{(y-y_2)^2} + 1}} \right)$$

$$0 = \frac{n_1}{c} \left(\frac{1}{\sqrt{\cot^2 \theta_1 + 1}} \right) - \frac{n_2}{c} \left(\frac{1}{\sqrt{\cot^2 \theta_2 + 1}} \right) = n_1 \sin \theta_1 - n_2 \sin \theta_2$$

$$\Rightarrow n_1 \sin \theta_1 = n_2 \sin \theta_2$$